

Part Four

CubeSat assembly, integration, testing and verification

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1 Introduction

The Kyushu Institute of Technology (Kyutech) inaugurated the Center for Nanosatellite Testing (CeNT) in 2010. CeNT is capable of performing all the tests necessary for nanosatellites whose size is up to $50 \times 50 \times 50$ cm and weight is up to 50 kg, except radiation. The test facility of CeNT is open to external users. Since its opening, CeNT has been involved in testing of more than 2/3 of the <50kg class of satellites developed in Japan. CeNT has also tested various satellites from abroad, such as satellites from Singapore, Thailand, Vietnam, the Philippine, Egypt, Costa Rica, etc. [Table 1](#) lists the CubeSats tested at CeNT, from year 2012 to June 2019. As of June 2019, CeNT tested 30 CubeSats ranging from 1 U to 6 U. Note that there are many other non-CubeSats tested at CeNT, not listed in [Table 1](#).

Based on the experience gained through the activities at CeNT, the author served as the project lead of ISO-19683, “Space systems—Design Qualification and Acceptance Tests of Small Spacecraft and Units,” which was published in 2017. Kyutech not only tests satellites from external users but also develops its own satellites. From 2012 to 2019, Kyutech launched 18 satellites, among which 14 were CubeSats. According to Bryce Space and Technology. [\[1\]](#), Kyutech is the number one academic operator of satellites, less than 600kg launched since 2012. Among 18 satellites, the author was the principal investigator of 16 satellites, 8 projects. In this chapter, based on this experience, a discussion of how to improve the reliability of CubeSats while keeping the nature of low cost and fast delivery, is presented. ISO-19683 will be described in detail.

2 Reliability growth through AITV activities

According to ISO-10795, “Space systems—Programme management and quality—Vocabulary,” “Assembly” is “combination of parts, components, and units that form

Table 1 List of CubeSat tested at CeNT, Kyutech (as of June 2019).

Launch year	Satellite name	Size	Organization
2012	FITSAT	1 U	Fukuoka Institute of Technology
2014	OPUSAT	1 U	Osaka Prefectural University
2014	KSAT-2	1 U	Kagoshima University
2014	INVADER	1 U	Tama Art University/University of Tokyo
2016	Stars-C	2 U	Shizuoka University
2017	Aoba VELOX-III (AV3)	2 U	Kyushu Institute of Technology/Nanayang Technological University (Singapore)
2017	Freedom	1 U	Tohoku University/Nakashimada Engineering Works
2017	TOKI	1 U	Kyushu Institute of Technology
2017	Ghanasat-1	1 U	Kyushu Institute of Technology/All Nations University College (Ghana)
2017	Mazaalai	1 U	Kyushu Institute of Technology/National University of Mongolia (Mongolia)
2017	BRC Onnesha	1 U	Kyushu Institute of Technology/Brac University (Bangladesh)
2017	Nigeria Edusat-1	1 U	Kyushu Institute of Technology/Federal University of Technology Akure (Nigeria)
2017	VELOX-II	6 U	Nanyang Technological University (Singapore)
2018	Irazu	1 U	Tecnológico de Costa Rica (Costa Rica)
2018	BHUTAN-1 (BIRDS-2)	1 U	Kyushu Institute of Technology/Department of Information and Telecom (Bhutan)
2018	MAYA-1 (BIRDS-2)	1 U	Kyushu Institute of Technology/Department of Science and Technology—Philippine Council for Industry, Energy and Emerging Technology Research and Development (the Philippine)
2018	UiTMSAT-1 (BIRDS-2)	1 U	Kyushu Institute of Technology/Universiti Teknologi MARA (Malaysia)
2018	SPATIUM-1	1 U	Kyushu Institute of Technology/Nanayang Technological University
2018	Stars-AO	1 U	Shizuoka University
2018	AUTcube2	1 U	Aichi University of Technology
2018	KnackSAT	1 U	King Mongkut's University of Technology North Bangkok (Thailand)
2019	Aoba VELOX-IV	2 U	Kyushu Institute of Technology/Nanayang Technological University (Singapore)
2019	Uguisu (BIRDS-3)	1 U	Kyushu Institute of Technology
2019	Raavana-1 (BIRDS-3)	1 U	Kyushu Institute of Technology/Arthur C Clarke Institute for Modern Technologies (Sri Lanka)

Table 1 Continued

Launch year	Satellite name	Size	Organization
2019	NepalSat-1 (BIRDS-3)	1 U	Kyushu Institute of Technology/ Nepal Academy of Science and Technology (Nepal)
—	BIRDS-4	1 U	Kyushu Institute of Technology/University of Philippine Diliman/Paraguay Space Agency
—	HSKSAT	3 U	Harada Seiki
—	Innosat-2	3 U	Astronautic Technology (M) Sdn Bhd (Malaysia)
—	NARSSCUBE-1	1 U	National Authority for Remote Sensing and Space Sciences (Egypt)
—	NARSSCUBE-2	1 U	National Authority for Remote Sensing and Space Sciences (Egypt)

a functional entity,” “Integration” is “process of physically and functionally combining lower-level products (hardware or software) to obtain a particular functional configuration,” and “Verification” is “confirmation through the provision of objective evidence that specified requirements have been fulfilled”. In other words, assembly is a process to combine satellite components (ISO-19683 uses the word of “unit” for “component”) to make a satellite system or subsystem. Before assembly, each component needs to be “verified” against its requirements. Integration is a process that occurs after the assembly to make the subsystem or the system function as whole. During the assembly and integration processes, the subsystem or the system needs to be “verified” against its higher level requirements. Testing is one verification method out of many verification methods, such as analysis, demonstration, inspection, or review of design (and verification by similarity). Testing can be categorized into environment tests, functional tests, and measurement tests. Although testing is not only the verification method, due to its significance, the system life cycle phase from completion of component development until launch is often referred to as AIT or AITV. It should be stressed, however, that component verification and testing is an important process that should occur before the system AITV, as a defect in a component can stop the entire AITV process if the premature component is integrated into the system without proper verification and testing.

The advantages of CubeSats are low cost and fast delivery. These advantages are gained by the extensive use of commercial-off-the-shelf (COTS) components, parts, and manufacturing methods. COTS components are not meant for use in space. Therefore the advantages are gained by sacrificing reliability of individual satellites against low cost and fast delivery. In fact, several statistics show the poor success rate of nano-satellites including CubeSats. Bouwmeester et al. [2] showed that only 48% of nano-satellite (defined by a weight of less than 10kg) succeeded in mission after the successful launch. The more recent survey by Swartwout et al. [3] shows that only 50% achieved partial or full mission success and 25% fell in dead-on-arrival (DOA) that meant no radio was heard from the satellite.

According to Saleh et al. [4], satellite reliability can be modeled by Weibull statistics. The reliability is defined as the probability of nonfailure from time zero (deployment into orbit) to a given time. The reliability shows a sharp decrease after the launch but approaches an asymptotic decay. The initial reliability decay can be attributed to infant mortality due to mismatch of satellite design or manufacturing against operational (space) environment. The later steady decay of reliability can be attributed to random failure of individual components/parts.

During AITV, a satellite system goes through a series of tests. The tests may detect various defects in design, material, workmanship, etc. Once a defect is detected, it is corrected, and the test resumes or the development process goes back to an earlier phase. The failure rate decreases due to improvement and keeps decreasing as the testing continues until it becomes a steady value indicating the shift to the random failure mode. This is so-called Duane model of reliability growth [5]. According to Duane [5], the failure rate, λ , has the following time dependence, while the reliability grows,

$$\lambda(t) = \frac{\beta}{\alpha^\beta} t^{\beta-1} \quad (1)$$

where t is the time. The meanings of α and β are explained later.

According to Crow [6], the failure is modeled by a non-homogeneous Poisson process, where the failure rate, λ , is not constant and decay as the time progresses until it reaches to a steady value. It is referred to as the reliability growth model. Based on the Poisson process, the probability that no failure occurs from time zero to t is given by the following:

$$R(t) = \exp \left(- \int_0^t \lambda(t') dt' \right) \quad (2)$$

We substitute Eq. (1) into Eq. (2) and integrate over the time. Then, we obtain

$$R(t) = \exp \left(- \left(\frac{t}{\alpha} \right)^\beta \right) \quad (3)$$

This is equal to the Weibull distribution with α as the scale parameter and β as the shape parameter. Fig. 1 shows an example of Weibull distribution. The smaller the scale parameter α , the failure rate is higher. The smaller the shape parameter β , the more failure will occur initially.

The fact that the failure of CubeSats in orbit is still modeled by Weibull distribution showing strong infant mortality means that the testing of CubeSats is not performed thoroughly to improve the reliability up to a point where the random failure of individual subsystem/unit/parts dominates.

We can regard the AITV processes as cumulating the time in Eqs. (1)–(3) before the satellite is launched. By cumulating the time the failure rate given by Eq. (1) decreases significantly already when the satellite is launched. Fig. 2 explains this point schematically.

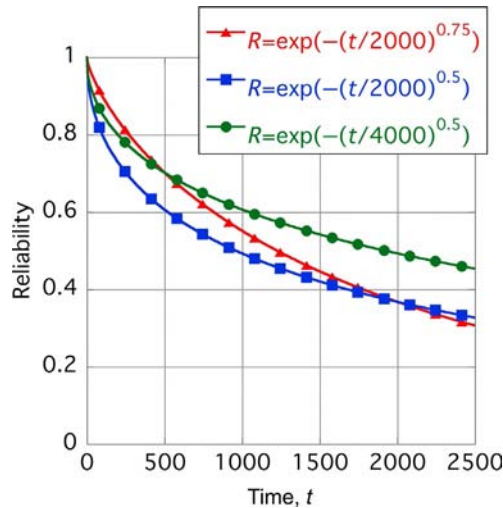


Fig. 1 Example of Weibull distribution.

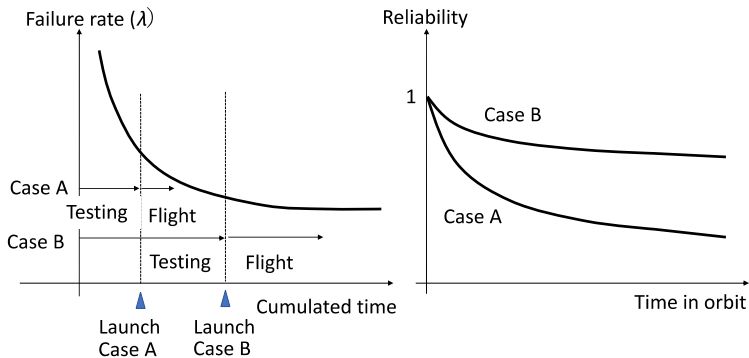


Fig. 2 Reliability growth and improvement of in-orbit reliability by AITV before launch.

Fig. 3 shows an example of the reliability growth observed in an actual satellite project [7]. We recorded the number of failures observed during AITV process of HORYU-IV that was developed at Kyushu Institute of Technology from 2013 to 2016. Although HORYU-IV was not a CubeSat, it was developed as a typical university-class satellite. Its size, $30 \times 30 \times 30$ cm, and mass, 10 kg, are equivalent to those of a 6 U CubeSat. The vertical axis of Fig. 3 is the cumulative number of failures observed during AITV process of an engineering model. The time zero is the time when we started to assemble various components into the satellite system. The horizontal axis is the total time spent in AITV. From the beginning to 180 h, numerous failures were observed during structure assembly and electrical performance test. After that the pace of finding new failures decreased. Fig. 3 clearly illustrates the

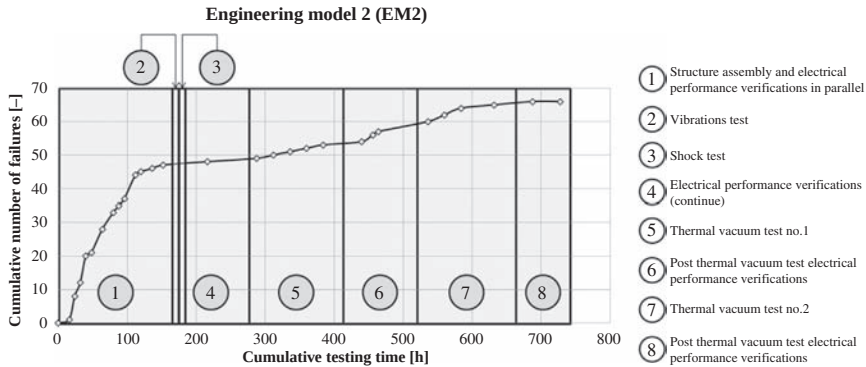


Fig. 3 HORYU-IV cumulative number of failures against cumulative testing time of Engineering Model [7].

importance of the AITV processes. Even if individual components work, once they are integrated into a system, there are always new problems found. Therefore, when we make the project schedule, a significant amount of time should be allocated to system AITV.

The question is how much time we should allocate to AITV and accumulate testing time to decrease the failure rate before the launch. The reliability certainly grows by cumulating more testing time on the ground. But its growth rate decreases as we accumulate over time. When to stop testing and launching is a very difficult question and has no clear answer. Perhaps, it is determined based on the experience of team members. Based on a 2016 study by Swartout et al. [3], the success rate of the second satellite in university CubeSat projects clearly increases. This increase is described by a simple learning curve, although most likely students who are working on the second satellite are different from those worked on the first satellite. For a constellation program the satellite in-orbit reliability increases generation by generation as improvements are implemented based on the on-orbit results. Although we cannot say how much testing is enough, we can say how much testing is necessary. The international standard ISO-19683 provides the minimum testing criteria. In the next section, we look at ISO-19683 in depth.

3 International standard applied to CubeSat AITV activities

3.1 ISO-19683

An activity to make an international standard of nanosatellites that included CubeSats was initiated in year 2011. The standard aimed at answering several needs. The first and most significant one was to improve reliability of nanosatellites. It was clear that verification and testing processes needed improvement. There were already several testing standards available, such as ECSS-E-ST-10-03C [8], NASA-STD-7002A [9],

and GSFC-STD-7000 [10]. Even an ISO standard of satellite systems and unit testing existed as ISO-15864 [11]. But applying those standards to CubeSats was not a solution because of the increased cost and time necessary to implement them. There was not a consensus as to what tests were needed, if needed what test levels were appropriate, and what tests were not needed. There was a need for written guidelines related to testing nanosatellites to improve their reliability while keeping the nature of low cost and fast delivery.

The second need was related to trade studies leading to selection of nanosatellite components. In 2011 nanosatellite components were already widely traded on the Internet. To many nanosatellite developers, the components looked attractive in terms of price and the time to be saved by using the commercial product instead of building the components from scratch. There was little guarantee, however, associated with those products. The test history was not transparent to the buyers. Flight heritage did not guarantee that the products sold to the customers were made of the same parts that had been tested. The standards aimed at giving the minimum assurance that the products went through the known environment tests whose test conditions were well documented.

In May 2011 the author proposed the new standard at ISO/TC20/SC14 that encompasses all spacecraft hardware related standards. The first reaction from the community was “CubeSats are a students’ toy. They are polluting orbits with debris. Do something.” There was strong feeling against not only CubeSats but also all kinds of small satellites among the ISO committee members that were made mostly of people from space agencies and big satellite companies. A group was needed to support the initiative. Thanks to the Japanese government’s funding support, the author could host a workshop every year to discuss testing standards. Later, this group grew to discuss wider aspects of small satellites and led to an International Academy of Astronautics (IAA) study of “Definition and Requirements of Small Satellites Seeking Low Cost and Fast Delivery,” which proposed a concept of “lean satellite” in its final report published in 2017 [12]. The testing standard was finally published as ISO-19683 “Space systems—Design Qualification and Acceptance Tests of Small Spacecraft and Units” in July 2017. Although the title of ISO-19683 indicates “small spacecraft,” CubeSats are well within the scope of the standard. This section discusses CubeSats AITV processes based on the standard. Interested readers are encouraged to review the full ISO 19683 [12].

The scope of ISO-19683 is to “provide test methods and test requirements for design qualification and/or acceptance of small spacecraft or units. It provides the minimum test requirements and test methods to qualify the design and manufacturing methods of commercial small spacecraft and their units and to accept the final products.” “Small spacecraft” is used in the standard as a result of compromise made to have the consensus among the ISO member countries. The standard defines “small spacecraft” as one that “utilizes non-traditional risk-taking development and management approaches to achieve low cost and fast delivery with a small number of team. To achieve these two points, low cost and fast delivery, satellite design relies on the use of non-space-qualified commercial- off-the-shelf (COTS) units, making satellite size inherently smaller. The design accepts a certain level of risk associated with the use of COTS.” Fig. 4 shows the target of ISO-19683.

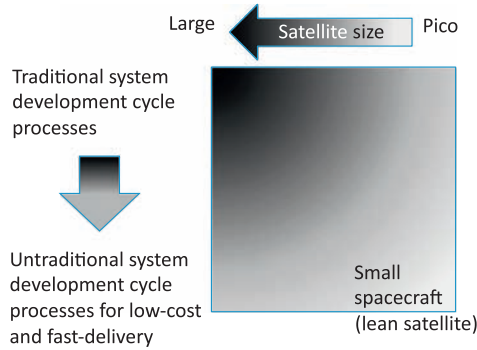


Fig. 4 Applicability of ISO-19683.

Although the scope indicates “commercial small spacecraft,” the document can be used for any kind of satellites such as academic and civil. The standard can be used as a guideline for the newcomers to the space sector, as a common document in an international satellite project, or for other purposes.

Satellite testing can be categorized into three types, qualification testing (QT), acceptance testing (AT) and protoflight testing (PFT). Generally speaking, QT is done to prove that the design meets the requirements. QT applies more environment stress, that is, margin, than the flight predicted level to a qualification model that is a non-flight item. AT is done to prove that the flight model meets the requirement, and it is free from latent workmanship errors and material defects. AT applies the maximum predicted environmental stress. PFT is done to save the cost of making the qualification model and doing QT. The model used for PFT is called a protoflight model that goes through the same environmental stress level, such as vibration amplitude, as QT for the same duration as AT. In this model the protoflight model is eventually deployed to orbit. ISO-19683 describes the requirements for qualification test (QT), acceptance test (AT), and protoflight test (PFT) at unit (component) level and system level. For system level test the meaning of QT, AT, and PFT are the same as the ones for traditional satellites, which is defined in ISO-15864. For unit level test the meaning of AT and PFT are the same as the ones for traditional satellites, which is defined in ISO-15864.

3.2 Satellite system tests

ISO-19683 provides a table of satellite system test items. The table identifies whether each test item is required, optional or not applicable for QT, AT, and PFT.

For QT, the required system tests are the following: electrical interface, functional test, mission test, electromagnetic compatibility (EMC) test, deployment test, physical property measurement test, launcher/spacecraft interface test, random vibration test, and thermal test.

For AT the required system tests are the following, electrical interface, functional test, mission test, deployment test, physical property measurement test,

launcher/spacecraft interface test, random vibration test, and end-to-end mission simulation test.

Between QT and AT the differences are EMC test and thermal test required for QT only and end-to-end mission simulation test for AT only. Annex-D of ISO-19683 provides the selection logic flows whether optional test items, such as sinusoidal vibration test, shock test, and thermal balance test, are necessary or not.

3.2.1 EMC test and end-to-end mission simulation test

The system electromagnetic compatibility (EMC) test is designed to demonstrate satellite's compatibility with the self-induced electromagnetic environment. To properly test for EMC, the satellite should be operated at full power. The test should satisfactorily demonstrate electrical and electronic equipment operation in conjunction with expected electromagnetic radiation and conduction from other internal units in various operational modes. In other words the test should demonstrate that the satellite is free from its own electromagnetic noise. There are several cases in which EMC test becomes important. One is related to reception of uplink data (command) from the ground. The signal from the ground is already very weak when it is received by the satellite, because of the long distance between the ground station and the satellite, so-called free space loss. In the case of 430 MHz UHF signal traveling 400 km distance, the free space path loss is 137 dB, which means the signal strength is weakened by more than 13th orders of magnitude from the original strength at the ground. A slight noise entering into the receiver may jeopardize the signal reception. Another example is an error in the sensor data reading when the noise is coupled to analogue data lines.

Lastly, when a satellite carries a device accompanied with switching of large current and/or high voltage, EMC is a strong concern. If the satellite carries electric propulsion system, especially a pulse discharge type, there is a significant EMC issue. The noise may often interrupt the functionality of digital electronics including microprocessor onboard. It is recommended, for example, to fire the thruster in a vacuum environment while the satellite operates free from external cables, including the power lines. The satellite components including the thruster head may be damaged. Therefore this test should be done at QT phase, and the test results can be reflected to the flight model. A CubeSat is a highly integrated system where so many coupling paths exist between the noise source and the device affected either by radiation or conduction. It is not practical to verify that the satellite is free from EMC issues via analysis. It is the best to demonstrate that the satellite system works in its fully assembled configuration even if all the internal components are working independently. An electromagnetic anechoic chamber may not be necessary as long as the external noise is negligible compared with the internally generated noise. When we need to characterize the minimum power level to succeed in uplink, it is better to perform the test in an anechoic chamber to avoid the external noise influence.

End-to-end tests are intended to verify that the satellite can be controlled and operated by the specified ground network. The emphasis is to verify the flight software onboard the satellite and communication between the satellite, the ground station,

and the ground data system. To verify the flight software, the tests should include simulations of all operational modes in nominal situations (e.g., early orbital operations, mission operations, and decommissioning) within the constraints of what can be simulated on the ground. The nominal operational modes should be tested long enough to detect any latent software error that manifests only after a certain run time. Many CubeSats do power cycling or resets to recover from single events or software hang-ups. The recovery mode from these events should be tested extensively. If there is a command of forced reset to be sent, the command should be sent from the ground station, and the satellite recovery should be confirmed by telemetry sent from the satellite to the ground station.

The compatibility of the RF airlink between the satellite and the ground network should be tested before launch. We can measure the lowest RF power level the satellite receiver or the ground station receiver can decode by directly connecting the receiver and the transmitter (or a signal generator) with variable attenuator inserted between the two. Using the measured value the link budget calculation can be closed with known (or measured) antenna gains. But this is an ideal case. In reality the effective antenna gain, especially the one onboard the satellite, may be different from the design value. The satellite antenna may pick up the radiated noise from the satellite itself. Considering these facts the final check should be done whether the RF air link between the satellite flight model with antennas deployed and the ground station is valid or not. If the license permits, it is recommended to perform the test in open air with a distance far enough to assure the far field of radio propagation. The open-air test has an advantage over the test in an anechoic chamber as the real ground station antenna can be used. [Fig. 5](#) shows a photograph of open-air long distance test of BIRDS-3 (Uguisu), where the satellite was placed at a mountain top of 622 m with the horizontal distance of 6370 m from the ground station at 54 m height.

3.2.2 Deployment test

The deployment test verifies that the deployment mechanisms, such as antennas and solar panels, can function according to the design requirements. Antennas and solar panels are the most common deployable components for CubeSats. Their deployment is often critical to mission success. Testing deployment repeatedly may be necessary if the deployment mechanism involves parts whose quality variation is uncertain. The number of repeated deployments should be determined from the viewpoint of securing enough confidence and the cost/schedule associated with the test. The test may be performed in atmosphere. The difference of environment such as vacuum, temperature, and microgravity should be properly assessed during test planning. The worst-case condition should be assumed. Many CubeSats use synthetic fiber thread to hold the deployable and cut the thread by a resistance heater, for example, nichrome wire. The worst condition is the case in which the heater is not heated to a high temperature high enough to cut the thread. To simulate that condition the deployment should be tested in the coldest temperature, but in an atmospheric condition so that the air can remove the heat away from the heater with the lowest battery voltage condition so that the current to the heater is the minimum. If one uses mechanical parts, such as



Fig. 5 Photograph of open-air long distance communication test. Actual setup at the mountain (top), closer view of satellite (bottom left) and view of ground station from the mountain (bottom right).

spring and hinge, to release the solar panel and latch after deployment, the worst condition should be tested in vacuum environment at the coldest temperature.

3.2.3 Launcher/spacecraft interface test

The Launcher/Spacecraft interface test is the so-called fit-check. The test simply involves inserting an assembled CubeSat into a POD to see whether it can fit into the POD and exit the POD smoothly when it is pushed by the spring at the bottom of the POD. The test should be done with a qualification model before the flight model structure is manufactured. It is common that a CubeSat does not fit into a POD in the first trial. The CubeSat standard indicates that the external dimension should be $100 \text{ mm} \pm 0.1 \text{ mm}$. Many machine shops can make CubeSat structural components within the tolerance limit of $\pm 0.1 \text{ mm}$. But, once the structure is assembled, slight distortion in three-dimension often prevents the satellite from fitting into the POD. If the satellite does not fit, the structure must be adjusted until it fits into the POD. It is not advised to perform such adjustment during the last minutes of flight model assembly. Therefore the fit test should be done as early as possible in the satellite project once the



Fig. 6 Photo of fit-check.

structure manufacturer is identified and the initial set of structural parts/components are delivered and assembled. Once the fitting is confirmed with a qualification model, it is advised not to change the structure design nor its manufacturer. [Fig. 6](#) shows a photograph taken during a fit-check test.

3.2.4 Thermal test

Thermal tests consist of thermal vacuum tests, thermal cycle functional tests, and functional tests in vacuum. Among the three tests, either thermal vacuum or a combination of thermal cycle function and functional test in vacuum are required. When there is a large temperature distribution within a satellite, thermal vacuum test should be chosen. Thermal vacuum testing is preferred because it is more flight representative. But not all CubeSat developers have access to the test facility as shown in [Fig. 7](#). The difficulty with this test is often the cooling capability of the test facility. Without a cooling capability, finding a vacuum chamber for testing is relatively easy, and even purchasing a chamber large enough to accommodate a CubeSat is not difficult as there is a market of second-hand vacuum chambers that sells unused chambers released from other terrestrial industries, such as semiconductors.

The purpose of the thermal vacuum test is as follows:

- (a) to check the performance at the high-temperature limit in vacuum,
- (b) to check the performance at the low-temperature limit in vacuum,
- (c) to check the susceptibility to the temperature cycle in vacuum,
- (d) to detect latent defects of workmanship and design.

The functional test in vacuum using a simple vacuum chamber without a cooling capability can fulfill item (a). Items (c) and (d) can be partially fulfilled by thermal cycling, although the test article is not in vacuum. The question is whether the performance test

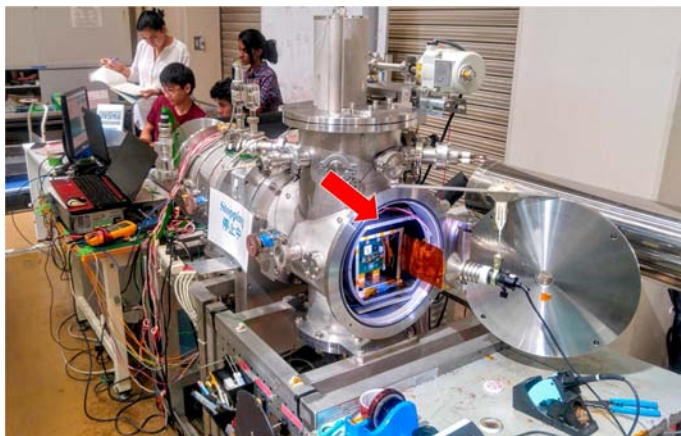


Fig. 7 Thermal vacuum test facility. There is jacket (pointed by a red (gray in print version) arrow) filled with liquid nitrogen to simulate the cold space environment.

at low temperature in a thermal cycle chamber, that is, atmospheric pressure, can fulfill item (b).

The largest difference between the two environments, vacuum chamber, and thermal chamber, is the presence of air (or nitrogen gas) convection. The convection tends to decrease the temperature difference within the satellite. During the system test the temperature difference should be properly simulated as it can cause a systematic defect that cannot be revealed by functional tests in uniform temperature.

Fig. 8 shows the temperature distribution of an electrical power control unit measured in vacuum and in atmospheric condition. The unit is operated in the same condition except the ambient environment. We see points where temperature is locally enhanced, where heat dissipation occurs as the power device efficiency is not 100%. The heat spots are more visible in the vacuum as air convection cannot remove the heat away. Although Fig. 8 is for an electrical power control unit for a 100 W-class satellite, a similar situation may arise in CubeSat at a DC/DC converter, a microprocessor, a RF amplifier, and others. If that is the concern, thermal vacuum test should be performed with the lowest and highest temperature extreme expected in orbit plus some margin. In ISO-19683 the QT level is defined as the design temperature and

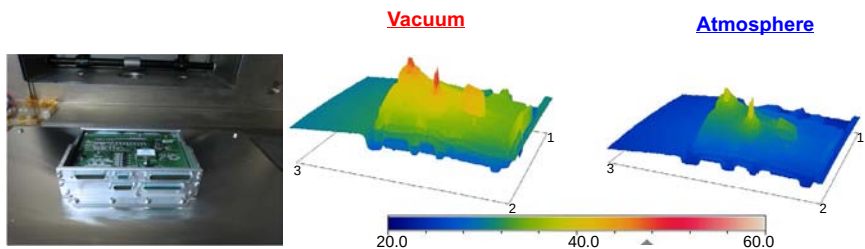


Fig. 8 Temperature distribution around electrical power system circuit board (left) in vacuum (center) and in atmosphere (right).

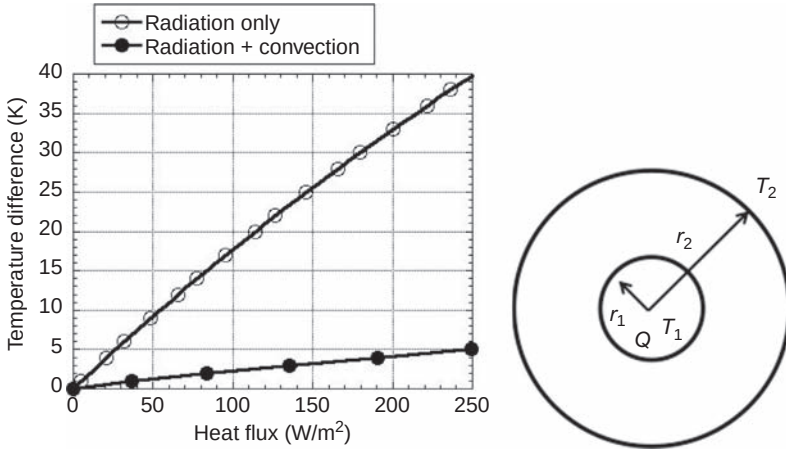


Fig. 9 Temperature difference between the outer sphere and inner sphere when there is heat transfer due to radiation only and due to radiation and air convection.

qualification margin (minimum $\pm 5^\circ C$) with two cycles or more. For traditional satellites the temperature margin in QT is often $10^\circ C$ or more [13].

Fig. 9 shows temperature differences calculated for concentric spheres. The inner sphere assumes an internal electric and electronics unit. The outer sphere assumes its surroundings, such as the satellite external surface or the mounting panel. From the inner sphere a heat of Q is transferred to the outer sphere. The heat flux is given by Q/A_1 , where A_1 is the surface of the inner sphere. Assuming a sphere of 10 cm diameter, $r_1 = 0.05$ m, 250 K as the outer sphere temperature, $\epsilon_1 = \epsilon_2 = 0.9$ as the emissivity, the temperature difference $T_1 - T_2$ was calculated for a given heat flux Q/A_1 . For $r_1 = 0.05$ m and $r_2 = 0.06$ m, the area is $A_1 = 0.0314$ m². Then $Q/A_1 = 50$ W/m² corresponds to $Q = 1.6$ W, which is comparable with typical 1 U CubeSat power dissipation. If we do the thermal test in atmosphere, the temperature difference is only 1 degree between the outer and inner surfaces. In vacuum the difference is almost 10 degrees. Studies [14, 15] argue that a difference of 10 degrees between the two environments (vacuum and atmosphere) should represent the criteria for exemption of thermal vacuum of electronics units. Therefore, perhaps, a 1 U CubeSat can be tested by thermal cycle. When we consider the heat transfer due to conduction, the temperature difference $T_1 - T_2$ decreases further. Then, when the power level, Q , is low, 2U or 3U CubeSat may be also tested by thermal cycle, but not good for a satellite larger than those. Even if thermal vacuum test is replaced by thermal cycle, the functional test in vacuum should be done at least once to make sure that the satellite works in a vacuum environment.

Thermal balance testing is often considered optional for LEO missions and may be required only when enough accuracy in thermal analysis is required or when high power devices or multiple payloads are supported. For LEO CubeSats the main purpose of thermal balance test is to provide the data necessary to verify the analytical thermal model or to verify the thermal control system. The data are designed to derive

the test conditions of subsequent thermal test, especially the lowest and the highest temperature limits. The CubeSat internal units are tightly packed into a small volume. Unless they are intentionally insulated, which is often true for battery but not for other subsystems, the internal units often have more or less uniform temperature except for hot spots inside the units. For a 1 U CubeSat, a three node analysis that account for external surfaces, internal units, and battery is often good enough. For a larger CubeSats with deployable solar panels and/or active attitude control, more sophisticated analysis is necessary. Although the combination of thermal balance test and thermal analysis can provide useful information to determine the thermal test conditions, it should be noted that there are many CubeSats already flying in low Earth orbit, and many of them look alike. More than 200 CubeSats have been launched from ISS since 2013. They are flying in the same orbit. Many CubeSats have similar thermal characteristics if they do not have deployables or significant power loads. The surface thermal optical properties are more or less the same as the surface is dominated by solar cell coverglass. Substituting the envelope of flight temperature data of the past satellites into the lowest and highest temperatures of the thermal test is one way to skip the thermal analysis and the thermal balance test. We can also verify the thermal analysis by analyzing the orbit conditions of the past satellites instead of doing the thermal balance test. The flight data should be used more widely. A database of flight temperature data of various satellites in various orbits may be helpful to many CubeSat projects.

3.3 Satellite unit tests

3.3.1 Unit QT in ISO-19683

In this section unit-level QTs are considered. ISO-19683 assumes that unit QTs are carried out by the unit manufacturer. For any type of subsystem units, functional test, physical property measurement, random vibration tests are required for QT. Except for structural units, either thermal vacuum test or a combination of thermal cycle functional test and functional test in vacuum is required. For an electrical and electronic unit, EMC test and thermal cycle endurance test are required. Total ionization dose and single event tests are optional. They can be done when a manufacturer wants to demonstrate radiation tolerance of the product. For antenna and solar array, deployment test is required if they are deployable units. As they are exposed to outer space, thermal cycle endurance test is required. Each is briefly discussed in the succeeding text. For batteries, leakage tests are required.

ISO-19683 also provides a table that lists the test level and duration for unit QTs. For TID test, 10kRad or higher total dose with 0.01 Gy/s or lower dose rate is given at the test conditions. For a typical CubeSat, the shielding effect of external panels is equivalent to aluminum of 1 mm thickness. We can calculate total dose in orbit using a tool like SPENVIS (www.spenvis.oma.be/). According to SPENVIS, a CubeSat in an orbit of 400km altitude and 51° inclination receives less than 2kRad in 1 year.

The random vibration test level is given as 13.3Grms higher between 20 and 2000Hz. The vibration should be applied once in each orthogonal axis for 1 min.

The test level was derived based on analysis of test data from various 50 kg-class satellites and basic research carried out. For a CubeSat, which is much lighter than 50 kg, the level of 13.3 Grms needs to be increased. In other words, if the CubeSat unit cannot survive this level of random vibration test, it won't survive the rocket launch. It should be noted that if the satellite is launched to ISS, the system random vibration level is 4.4 Grms, which is very low as the satellites are carried as a package inside the cargo vehicle. Then 13.3 Grms as unit QT level may provide margin.

For thermal vacuum testing of an internal unit, the temperature range is specified as -15°C to $+50^{\circ}\text{C}$. The number of cycles is two or more and the test should be done in pressure of 1×10^{-3} Pa or lower. For thermal cycle functional test of an internal unit, the temperature range and the number of cycles are the same as those of thermal vacuum, i.e. -15°C to $+50^{\circ}\text{C}$ and 2 cycles or more. At each cold and hot temperature limit, functional tests should be carried out. To derive the temperature ranges, the temperature ranges of internal units on-board various 50 kg-class satellites were investigated. The temperature range of -15°C to $+50^{\circ}\text{C}$ corresponds to the coldest and the hottest case of the collected data. If only one cycle is done, there may be some singularities. Therefore, at least two cycles are required. Other standards require four or eight cycles.

The thermal cycle endurance test demonstrates the ability of the test article to withstand the stress imposed by thermal cycles in orbit. A functional test and visual inspection should be performed before and after the cycles. There is no need of doing the functional test during the cycles. The test level is -70°C (or lower) to $+100^{\circ}\text{C}$ (or higher) for an external unit such as solar array and -25°C (or lower) to $+60^{\circ}\text{C}$ (or higher) for an internal unit. The number of cycles is 24. The temperature range of -70°C to $+100^{\circ}\text{C}$ corresponds to a typical temperature range of solar panels at SSO. For an internal unit, the temperature range of -25°C to $+60^{\circ}\text{C}$ is the non-operational temperature range. It was derived by adding 10°C margin to the operational temperature range of -15°C to $+50^{\circ}\text{C}$.

3.3.2 Battery tests

When we test batteries, we need to consider not only whether the battery functions in space but also whether the battery comply with the safety requirement imposed by the launcher. The major concern about battery safety is leakage and rupture. It is required to expose the battery to a vacuum to check the leakage. Functions to control the hazard leading to rupture need to be tested before and after environmental tests.

When the battery packs are assembled by the satellite developer, screening of individual battery cells may be required. Fig. 10 show the process of battery cell screening required for ISS release. We took an example of BIRDS-2 that used a NiMH battery (Eneloop, HR-3 UPT). To verify the individual cells, we followed the flow in Fig. 10. First the battery characteristics were measured for each cell. They were open voltage, battery discharge capacity, mass, and charge and discharge profile along with the temperature, visual inspection, and odor. Among these the battery capacity measurement needed one cycle of charging/discharging test using a DC power supply and an electronic load. Then the batteries were exposed to 1×10^{-3} Pa vacuum for 6 h. Then the

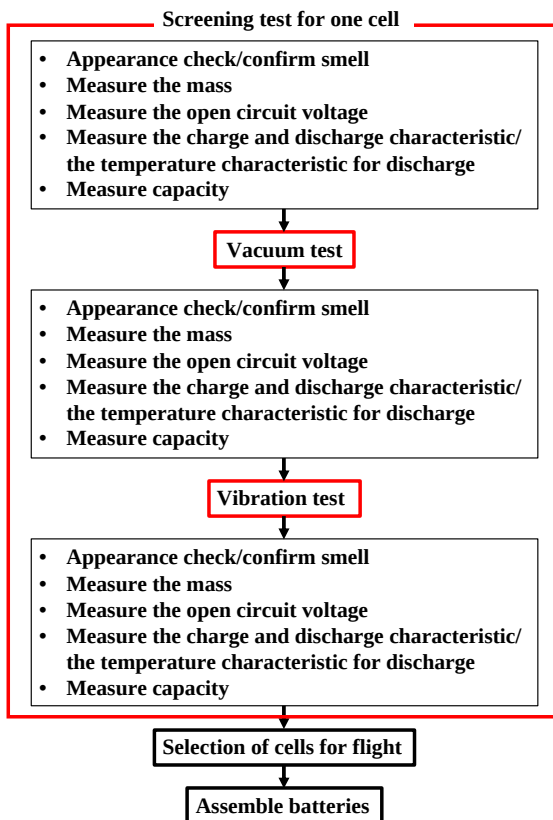


Fig. 10 Flow of battery screening of BIRDS-2.

characteristic measurement was done again to verify the change of open circuit voltage (less than 0.1%), the battery capacity (less than 5%) and the mass (less than 0.1%) to be within an acceptable range. Then, a random vibration of 6.1 Grms was applied for 1 min to each axis. Cells whose characteristics (open circuit voltage, capacity and mass) change was within the acceptable range were selected as flight batteries and assembled into a battery pack.

If the battery itself has a protection circuit against overcharging, overdischarging and external, short circuit, there is a need to demonstrate that the protection circuit works. The external short-circuit protection, that is, over-current protection, was confirmed by observing that the discharge stops once the current reaches a certain value, as shown in Fig. 11, which was taken from the test data of Aoba VELOX-III (AV3). These tests are very harsh for each battery. Therefore the tests were done using a battery from the same lot as the flight cells, not a flight cell itself.

Nowadays, many CubeSat developers purchase batteries from a component vendor that sells the battery already integrated into an EPS board. If the COTS EPS board is

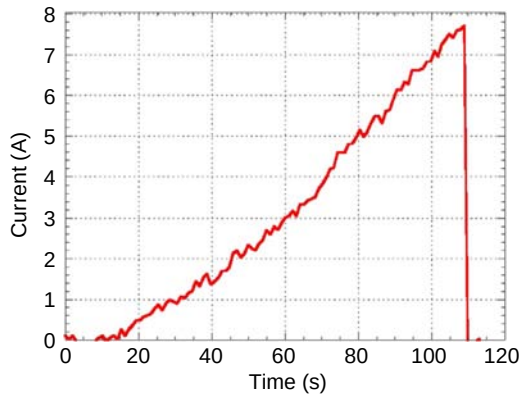


Fig. 11 Demonstration of overcurrent protection of AV3 battery.

already designed and verified regarding the safety requirements, procuring the COTS EPS instead of developing in-house saves significant amount of time in the safety management. But in many cases, it is not the case.

Kyutech assisted in the safety design and verification of Irazu, which was the first Costa Rican satellite released from the ISS in May 2018 (<https://directory.eoportal.org/web/eoportal/satellite-missions/i/irazu>). Irazu used a commercial EPS that had an integrated Li-ion battery. If the same principle as that for BIRDS-2 and AV3 was applied, each battery cell should have gone through environment tests, and the characteristics should have been measured. As the battery cell was already integrated into the COTS EPS, it was not possible to evaluate the individual cells. Therefore the safety verification of individual cells was done according to the JDX-2017078 “Guideline for safety design and verification for a small satellite using JEM Small Satellite Orbital Deployer with satellite-kit battery cell and EPS.” The battery test report (characteristics measurement, such as open circuit voltage, capacity, and mass before and after vacuum exposure and vibration) conducted by the manufacturer before shipment was used to verify the flight battery cell. Regarding overcharging, overdischarging, and external short circuit, the test data of the battery cells of the same lot were submitted by the manufacturer. The product serial number and the serial number listed in the document submitted by the manufacturer had to clearly show that the batteries are from the same lot. The documents provided by the manufacturer were substituted for the test reports to be otherwise conducted by the satellite integrator. The flight battery pack was verified through the system tests after it was integrated into the satellite system. The tests were vacuum exposure and vibration. The vacuum exposure test was done as part of the thermal vacuum test. The vibration test was done as a part of a protoflight model (PFM) vibration test (6.4 Grms, 1 min). Before and after each test and the charging and discharging characteristics of the battery pack were measured.

4 Concluding remarks

CubeSat AITV is a critical process intended to improve the satellite reliability and achieve the mission success. ISO-19683 was published in 2017 to provide the minimum set of CubeSat testing requirements. The standard is a living document. According to ISO regulation, each standard should be reviewed every 5 years. When ISO-19683 was published, a massive constellation was still on the horizon. The standard targeted mostly single satellite projects. One advantage of the CubeSat is its rapid speed of generational change. By experiencing multiple satellite projects and learning lessons from in-orbit results, we can improve ourselves and the mission success rate increases as the generation goes on. It is certainly true for a constellation program. Later generations go through the minimum set of testing on the ground, yet their mission success rate increases. A university project who had a satellite dead on arrival during its first mission often succeeded in getting signals from the satellite in the second mission and achieved the full mission success later. Traditional testing standard, such as ISO-15864 and ECSS also evolved after series of success and failure in early days of space exploration. The AITV processes based on the standards are very mature now and the mission success rate of traditional satellites is high. CubeSat testing standards also can evolve by collecting experiences of the community. For commercial satellite programs, it may be difficult to share the experience and data due to its proprietary nature. But there is little difficulty for university satellite programs to share the information. Looking at the evolving speed of the CubeSat community, if there is an appropriate platform to share the information, it may not be so difficult to have more mature CubeSat testing standards and have drastic improvement in mission success rate in very near future.

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